Tier 3 results — research

Items 39–46 of TASKS.md, executed 2026-10-12. Related work was done **first**, as item 44 instructs, and it changed the framing before a single experiment ran.

The short version

Tier 3 was meant to find a headline. Its first pass concluded that three of four research angles did not survive contact with the data. **A follow-up causal experiment overturned that for N1**, which is now the strongest finding in the project — see the correction under item 39. The other three verdicts stand.

#	Angle	Outcome
39	N1 plan drift	**SUPERSEDED — see the correction below.** Causal measurement makes it the **dominant** term (70.5%), not subordinate
40	N2 elasticity as descriptor	**Refuted** in Tier 1 — better description, no predictive gain
41	N3 budgeted cold start	**Works, and inverts the premise**: one well-chosen query beats twelve random ones
42	N4 conformal risk control	**Machinery works, guarantee voids** on unseen hardware (16/18 budgets missed)
43	N5 hardware embeddings	**Blocked** — needs 30+ machines
44–46	Related work	**Done**, and corrected four of eight citations plus found the closest prior work, which the list had missed

The through-line: **every angle fails for the same reason** — five machines, and a hardware signature that is effectively one-dimensional. That is now a finding supported from five independent directions, not a limitation to apologise for.

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39 — N1: plan drift

The question. The project assumes workload cost is hardware-invariant and only capability changes. For a relational optimiser that is false: PostgreSQL reads hardware-derived settings and picks a different plan, so the dev-side cost vector describes a plan production will not run. Nobody has decomposed cross-environment error this way (`RELATED_WORK.md`).

The design. Measured on **one** machine, varying only the planner's hardware *beliefs* (`max_parallel_workers_per_gather`, `effective_cache_size`, `work_mem`, `random_page_cost`). Hardware capability is constant by construction, so every runtime change is plan divergence with **no capability confound** — an isolation two machines cannot give, since there both terms move together. 22 TPC-H queries × 6 environments, 264 measurements.

Result 1 — drift is pervasive. 20 of 22 queries change plan; median 3 distinct plans, max 6. By single axis, from the baseline:

hardware belief changed	plans changed	runtime ratio when plan changed
`max_parallel_workers_per_gather` 2 → 8	**12/22 (55%)**	1.27×
`random_page_cost` 4.0 → 1.1	9/22 (41%)	1.39×
`work_mem` 4MB → 64MB	4/22 (18%)	2.74×

Core count is the dominant plan-changer — independent corroboration of the parallelism mechanism, from a completely different measurement.

Result 2 — but it is the smaller term. Splitting all environment pairs by whether the plan signature was identical: same-plan pairs swing runtime **2.16×**, different-plan pairs **2.49×** (Mann-Whitney $p = 1.7 \times 10^{-3}$). So only **~16%** of cross-environment variation is attributable to the plan changing. Most of it is the settings' effect on *execution*.

Result 3 — and the sign is backwards. Plan-volatile queries are predicted **better**, not worse: per-query error correlates with the number of distinct DOP levels at **r = −0.551, p = 0.010**. So volatility cannot be the mechanism behind the error. The coherent reading: a query whose plan changes with core count is a query the planner can *parallelise*, and parallelisable queries are exactly what the multi-core compute ratio captures.

Verdict at the time: N1 should not be the headline. That verdict is now **wrong**, and the correction is below.

CORRECTION (2026-10-13): N1 *is* the headline

The observational estimate above was diluted. A causal decomposition using forced plans (`results/tier4_plan_drift_causal.md`) finds that **when the plan actually changes, plan divergence accounts for 70.5% of the runtime difference** — the pure plan effect is 1.673x against 1.145x for the pure settings effect, over 30 cells where forcing verified.

Both numbers are correct; they condition differently. The ~16% above averages over **all** environment pairs, including the majority where both environments chose the *same* plan and a plan effect cannot exist. The 70.5% conditions on the cells where the plan **did** change.

Put together: **the plan changes often (20/22 queries; 55% when core count alone moves), and when it changes it explains about 70% of the runtime difference.** Plan divergence is a first-order mechanism, not a refinement. The research question it implies — *which plan will production choose* rather than *how long will this plan take there* — is both more tractable (the planner is deterministic) and unaddressed in the literature.

One caveat does real damage to the precision of the 70.5%: forcing verified on only **41%** of differing-plan cells, and the failures are not random — they are plans whose shape `pg_hint_plan` cannot express. The estimate is over a biased, plausibly simpler subsample.

The inverted correlation (r = −0.551) still stands and is still interesting: plan-volatile queries are predicted *better*, because volatility marks parallelisability.

41 — N3: budgeted cold start

Post-calibration MAPE, leave-one-machine-out, scored only on queries **not** revealed:

k	random	spread	dopt	elasticity	oracle
1	19.51	**13.20**	14.45	13.20	12.70
3	15.82	12.57	13.78	15.82	12.44
12	13.45	12.25	13.22	14.05	5.76

Uncalibrated baseline: **15.84%**.

- **One well-chosen measurement beats twelve arbitrary ones.** `spread` at k=1 (13.20%) beats `random` at k=12 (13.45%). *Which* queries you run matters more than how many — the opposite of `addition7_online_loop.py`'s arbitrary 12, and a far cheaper operating point.
- **Calibrating from a single random query makes things worse** (19.51% vs 15.84% uncalibrated). One query is a noisy estimate of the machine correction, and a bad correction is worse than none. A loop revealing queries in arbitrary order can *degrade* the predictor before improving it — undetectable in the previous design, which never measured k=1.
- `elasticity` **and** `dopt` **get worse as k grows.** Both chase *extremes*; for estimating a single scalar you want *representative* queries. `spread` in predicted-scaling space — the only objective aligned with the quantity being estimated — is the most stable column.
- **The gain is modest overall** (15.84% → ~12%), consistent with Tier 1: the analytical formula is already close to per-machine unbiased, so little systematic correction remains. Worth doing because it is nearly free, not because it transforms anything.

42 — N4: conformal risk control

A deployment owner states a budget — "no more than 5% of what I approve may breach" — and the gate threshold `λ` is *derived* by conformal risk control rather than chosen.

The machinery works. The guarantee does not survive an unseen machine: 16 of 18 configurations exceed their stated budget on the held-out machine, in the worst cases by several times. At τ=1.5 with a 5% budget, `λ=1.00` approves 92.5% of candidates and realises 8.33% risk, worst fold 26.8%.

This is not an implementation defect. CRC guarantees its bound **under exchangeability**, and a held-out machine is not exchangeable with the training machines — the c5a finding again. Tier 1 reached the same conclusion from the coverage side (97.0% marginal at nominal 95%, but 90.5% on the worst machine). Two independent formulations, same answer.

The gap is exchangeability, which is a data problem, not a method problem. Keep the framing — it beats the shipped gate's hand-set 0.6, which Tier 1 showed was not even estimable at n=5 — but report a stated risk budget on unseen hardware as *aspirational*, and lean on abstention instead.

43 — N5: blocked

Learned hardware embeddings need 30+ machines to factorise a machine × query matrix. Tier 1 already answered the underlying question negatively by a cheaper route: with two axes correlated at r = +0.958 and 97.9% of variance on one component, there is no second dimension to embed. Revisit after Tier 2.

44–46 — Related work

Full detail in RELATED_WORK.md. Two things worth surfacing here.

Four of eight citations were wrong or incomplete. "QPPNet" is not the paper's own name; the covariate-shift conformal paper is Tibshirani, **Barber, Candès & Ramdas** (three co-authors were being dropped); Ganapathi's title and Neo's eight-author list were off; the Roofline volume/issue still needs confirming.

The list missed the closest prior work. Heinrich, Binnig, Kornmayer & Luthra, **COSTREAM** (ICDE 2024) is a learned cost model taking hardware as input that claims generalization to unseen hardware. I initially recorded it as interpolation-only; that was wrong — its Exp 4 explicitly extrapolates beyond the training range and largely holds up. **So "no prior work evaluates cross-hardware extrapolation" must not be claimed.** Note Carsten Binnig co-authors both COSTREAM and the zero-shot cost models paper: the Darmstadt/DFKI group owns both nearest pieces of prior art and will plausibly review this.

What survives as novel, and only this:

1. **Relational OLAP rather than streaming** — and the distinction is substantive, because a relational optimiser chooses a different plan on different hardware, which is item 39's question. Streaming operator placement has no plan enumerator.
2. **Latency percentiles as the decision variable.** Every paper in the field predicts a point cost and reports *q-error* percentiles. None predicts p95/p99 of the latency distribution, which is what an SLA is written against.
3. **Abstention with calibrated coverage.** No prior work declines to answer.

What Tier 3 implies

The paper to write is a measurement study, not a better predictor. Every angle above failed in the same direction, and the failures are coherent and mutually corroborating:

- The hardware signature is one-dimensional (r = +0.958), so no weighting scheme can help — confirmed independently by the classifier ablation, the elasticity experiment, and N5 being moot.
- The compute signature is a hardware **× configuration** artifact: probed at `nproc` while a query can use `1 + max_parallel_workers_per_gather`. Invisible at 2–4 cores, catastrophic at
- **Every paper encoding hardware capability as a scalar is exposed to this, and none discusses it.**
- Exchangeability across machines fails, so both coverage and risk guarantees void on unseen hardware — shown twice, from independent formulations.
- Plan drift is real, driven by core count, and its *inverted* correlation with error is itself informative.

That critique generalises beyond CETP, which is what makes it publishable. COSTREAM uses four hardware features and never tests their independence either.

Target the honest venue. aiDM or DBTest at SIGMOD, where a rigorous negative-result measurement study with a public dataset is in scope. Not a VLDB full paper claiming state-of-the-art, which this evaluation cannot support and should not attempt.